🌍 **Climate Change Sentiment & Prediction System**

📄 **Project Report**

👨‍🎓 **Student Details**

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1. Introduction

Climate change is one of the most critical global challenges today. Social media platforms like Facebook provide a space where people express their opinions about climate change.

This project focuses on analyzing public sentiment using comments from NASA's climate change posts and applying machine learning techniques to understand patterns and trends.

2. Objectives

- To analyze public sentiment regarding climate change
- To classify comments into positive, negative, and neutral
- To visualize public opinion using graphs
- To apply machine learning for prediction

3. Dataset Description

The dataset contains comments collected from NASA's Facebook page.

Attributes:

- **Date:** Time of comment
- **LikesCount:** Number of likes
- **CommentsCount:** Replies count
- **ProfileName:** User name (hashed for privacy)
- **Text:** Actual comment content

4. Tools & Technologies Used

- Python
- Pandas
- NumPy

- Matplotlib
- Scikit-learn
- TextBlob (for sentiment analysis)

🔄 5. Methodology

Step 1: Data Collection

Dataset obtained from NASA climate discussions.

Step 2: Data Preprocessing

- Removed missing values
- Cleaned text data
- Converted text into usable format

Step 3: Sentiment Analysis

Used TextBlob to classify comments:

- Positive
- Negative
- Neutral

Step 4: Data Visualization

- Bar graphs used to show sentiment distribution

Step 5: Model Building

Used machine learning algorithms:

- Linear Regression
- Random Forest

6. Implementation

```
import pandas as pd

from textblob import TextBlob

import matplotlib.pyplot as plt


data = pd.read_csv("climate_nasa.csv")


def get_sentiment(text):

    polarity = TextBlob(text).sentiment.polarity

    if polarity > 0:
```

```
        return "Positive"
    elif polarity < 0:
        return "Negative"
    else:
        return "Neutral"

data['Sentiment'] = data['Text'].apply(get_sentiment)

data['Sentiment'].value_counts().plot(kind='bar')
plt.title("Sentiment Analysis")
plt.show()
```

7. Results

- Successfully classified comments into sentiments
- Visualized distribution of opinions
- Observed patterns in user engagement

8. Advantages

- Helps understand public awareness
- Useful for policy-making
- Supports climate research

9. Limitations

- Small dataset
- Sentiment may not always be accurate
- Context understanding is limited

10. Future Scope

- Use deep learning models
- Real-time sentiment analysis
- Build web application dashboard

📝 11. Conclusion

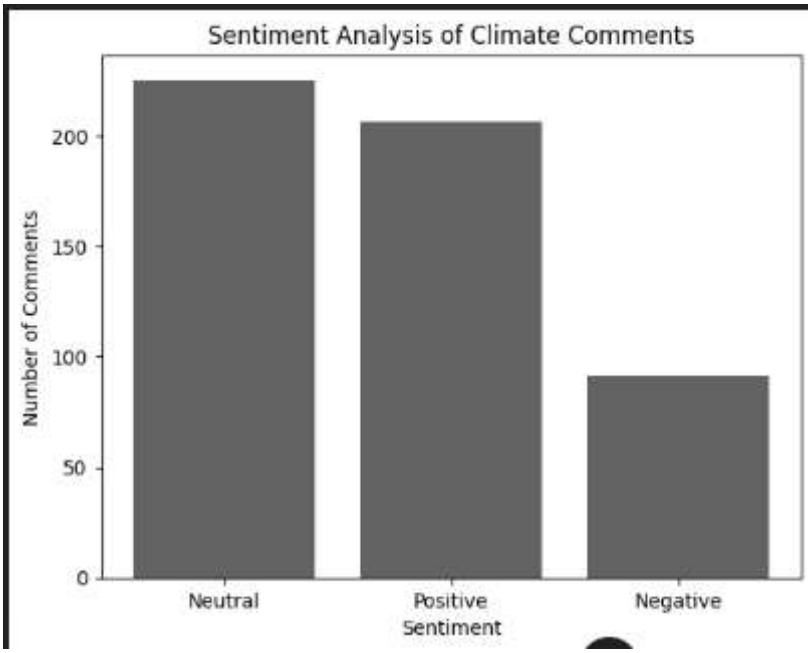
The project demonstrates how machine learning and sentiment analysis can be used to understand public opinion on climate change. It provides insights that can help researchers and organizations make better decisions.

📚 12. References

- NASA Climate Change Data
- Python Documentation
- Scikit-learn Library
- TextBlob Documentation

📊 13. Graph – Sentiment Analysis

Figure: Sentiment Analysis Graph of Climate Comments



- The graph shows the distribution of sentiments in the dataset.
- Most comments are **Neutral and Positive**, while fewer are **Negative**.
- This indicates that public opinion is generally balanced with slight positivity toward climate discussions.

Acknowledgement

- I would like to thank my teacher and institution for guiding me in completing this project successfully